Faculty of Science and Technology

Programme Structure & Syllabus

For

First Year

Bachelor of Technology (Computer Science and Engineering)

Bachelor of Technology (CSE-AIML)

Bachelor of Technology (CSE-Cyber Security)

Bachelor of Technology AIDS

Proposed Programme Code: BTECES

Pattern 2026

With Effect from Academic Year 2026 – 2027

	Authority	Date
Proposed by	Board of Studies in Engineering Sciences	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS **Dean of Faculty** **Director, IQAC**

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of Computer Engineering	
To create an intellectual and academically rich environment for careers in Computing Education and Research to fulfill global needs.	
Mission of the Department of Computer Engineering	
M1	To impart modern computing education through novel methods and research.
M2	To prepare the students for life-long learning in pursuit of excellence.
M3	To create technically competent human in socio-economic domain.
M4	To promote aspiring ethically conscious engineers demonstrating sustainable entrepreneurship and professional maturity in social context.

Values of the Department of Computer Engineering	
Teamwork.	Co-operation.
Understanding.	Integrity.

Programme Educational Objectives (PEOs)

PEO No.	Statement	
PEO1	**Preparation**	To prepare the students to evolve into a professional and committed data science technology workforce, by providing them with a global educational platform with innovative practices.
PEO2	**Core competence**	To provide a sufficient mathematical, statistical and computing theory knowledge base, leading to practical and long lasting data science solutions.
PEO3	**Breadth**	To ingrain problem solving skills in data science and sound engineering principles in students with the help of modern data science tools and techniques.
PEO4	**Professionalism**	To impart in the students professional and ethical practices based on standard guidelines as applied to the field of artificial intelligence to acquire the right attitude and aptitude for the benefit of society.
PEO5	**Learning Environment**	To guide students for higher education and lead them to avenues of entrepreneurship in areas of artificial intelligence and data science.

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3	PEO4
M1	1		1	
M2	1	3	3	2
M3	1			3
M4	2	2		

Affinity Level: 1- Slight, 2- Moderate,3-Substantial

Programme Outcomes (POs)

PO No.	Statement	
	Graduate Attributes	**Programme Outcome**
PO1	**GA 1:** Engineering Knowledge	Apply the knowledge of engineering, data science fundamentals and artificial intelligence to solve complex engineering problems.
PO2	**GA 2:** Problem Analysis	Identify, formulate and analyze complex engineering problems to create solutions using the first principles of mathematics, statistics, and data science.
PO3	**GA 3:** Design/Development of solution	Design solutions for system components that meet the specified needs for the benefit of society.
PO4	**GA 4:** Conduct investigation of Complex Problems	Apply research methods including design of experiments, statistical analysis and business interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**GA 5:** Modern Tool Usage	Apply appropriate data mining techniques and statistical tools to model complex engineering activities.
PO6	**GA 6:** The Engineer and Society	Apply reasoning using contextual knowledge to assess the needs of society and understand the responsibilities relevant to the engineer specializing in data science.
PO7	**GA 7:** Environment and Sustainability	Understand the impact of artificial intelligence as applied to the environment, and demonstrate the knowledge for sustainable development.
PO8	**GA 8:** Ethics	Adopt ethical practices as applied to the professional world to execute responsibilities and adhere to norms of the engineering profession.
PO9	**GA 9:** Individual and Team Work	Execute professional functions effectively as an individual, as well as a leader or member in diverse multidisciplinary teams.
PO10	**GA 10:** Communication	Communicate effectively with the engineering community and with society in solving complex artificial intelligence problems in terms of being able to comprehend and write effective reports, make effective presentations, as well as execute and receive clear instructions.
PO11	**GA 11:** Project Management and Finance	Demonstrate an ability to lead projects and build cost models in an interdisciplinary professional setting.
PO12	**GA 12:** Lifelong Learning	Develop learning mechanisms and inculcate the ability to prepare for lifelong learning in the context of technological change.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PEO1	1	1			2	3		1			1	2
PEO2	1	1	2						1	2		1
PEO3	1		2	3	1	1				2		2
PEO4	1						1					2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Imbibe foundational knowledge of computing theory, statistics, algorithms and programming paradigms to create innovative, scientific solutions for challenges in the field of data science and artificial intelligence.
PSO2	Prepare for recent trends in the data science industry by adapting to various computing platforms, architectures and changing software paradigms.
PSO3	Evolve new design ideas for real world problems in data science using appropriate mechanisms, design patterns, modeling methods and modern machine learning and statistical tools.
PSO4	Inculcate research attitude and aptitude in students using techniques and standards as applied to data science technology migration and evolution.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3	PSO4
PEO1	1		1	
PEO2	1		2	3
PEO3	1	2		2
PEO4	2			

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Structure

Faculty	***Science and Technology***	**Pattern (AY-2026-2027)**	2026
Department	**Engineering Science**	**Date (w.e.f.)**	
Programme	***BTech* Computer Science and Engineering, CSE-AIML, CSE-CS, AIDS**		

SEM-I

Course Type / NEP Type	**Tentative Course Code**	**Course Name**	**Teaching Scheme (Hours/Week)**			**Credit**	**Total Marks**
			L	**T**	**P**	**C**	
BSC	BTECES26101	Statistics and Differential Calculus	3	1	0	4	100
ESC	BTECES26102	Fundamentals of Electronics	3	0	2	4	100
ESC	BTECES26103	Computer Programming	3	0	2	4	100
VSEC	BTECES26104	Web Application Development	1	0	2	2	100
BSC	BTECES26105	Applied Science and Engineering Project I	0	0	4	2	100
CC	BTECES26106	Value Added Career Orientation I	2	0	0	2	100
IKS	IKS25AI01A	Indian Knowledge System	2	0	0	2	100
AU	BTECES26107	Induction Training			1*	0	
Total			**14**	**01**	**10**	**20**	**700**

Head of the Department **Dean**

---

*ERP to provide document upload facility for evidencing – BoS meeting MoM, AC MoM, final approved grid with BoS chairman signature, etc.

BTECES26101: Statistics and Differential Calculus

Course Type	BSC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	03Hr./Week	Lecture	03	CIE Marks	60
Tutorial	01Hr./Week	Tutorial	01	ESE Marks	40
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	04Hr./Week	Total	04		

Course Description

This course develops a foundation in probability, statistics, and differential models with applications in artificial intelligence, data science, and cyber security. Students explore random variables, statistical inference, and lightweight differential equations along with an introduction partial differentiation. The focus is on computational understanding, real-world applications, and R-based implementation.

Course Outcomes

CO No.	At the end of this course students will be able to …..
1	Apply probability concepts to model uncertainty in engineering and data-driven contexts
2	Analyze datasets using statistical measures and visualization tools (including R)
3	Perform and interpret hypothesis testing for decision-making
4	Formulate and solve ordinary differential equation models for engineering systems
5	Apply partial derivatives and multivariable calculus concepts in optimization and system analysis

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PS O3	PSO 4	
CO1	2	2	1	2	1							1	2		1	1	1,2,3
CO2	2	2		1	1							1	2		1	1	1,2,3
CO3	3	3	1	2	1	1						1	2		1	1	1,2,3, 4
CO4	3	3	1	1		1						1	1			1	1,2,3, 4
CO5	2	2										1	1				1,2,3, 4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom’s Taxonomy Level

SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)	
How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?	● Introduction to AI-supported statistical and mathematical tools at FYBTech level. ● Use of R, spreadsheets, and visualization tools for data analysis and graph plotting. ● Simple case studies on regression, probability, stock trends, and mathematical modeling. ● AI-assisted exploration of distributions, curve fitting, and differential equation applications. ● Focus on conceptual understanding first, followed by guided AI-based exploration.
SECTION B: AI Integration in Teaching & Learning	
How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?	● AI used only as a supportive learning aid, not as a replacement for teaching fundamentals. ● Faculty-led explanation of concepts, derivations, and problem-solving methods. ● Use of AI tools for visualization, practice questions, and revision support. ● Encouragement of critical thinking and verification of AI-generated outputs. ● Ethical and responsible use of AI emphasized throughout the course.
SECTION C: AI Integration in Assessment & Evaluation	
How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?	● Assessments include written tests, assignments, quizzes, and practical activities. ● Use of AI tools for data analysis and visualization tasks. ● Higher weightage given to manual calculations, interpretation, and conceptual clarity. ● Faculty moderation ensures fairness, originality, and transparency. ● Student performance measured through continuous assessment and CO attainment.

COURSE CONTENT

UNIT NO.01	DESCRIPTIVE STATISTICS	08 Hours	CO	BT L
Definition, Scope of Statistics, Collection and classification of data; different types of diagrams to represent statistical data; frequency distribution and related graphs and charts. **Central tendency:** Its measure and its uses. **Dispersion:** Its measure and their uses, **Moments:** Skewness and Kurtosis Statistical Analysis using R software in case studies			1	1,2,3
UNIT NO.02	**CORRELATION, REGRESSION & CURVE FITTING**	**08 Hours**	2	1,2,3
Scatter diagram, Correlation coefficient with its properties and limitations; linear regression and its uses., the principle of least squares, fitting of polynomials. Analysis of real data set using R.				
UNIT NO.03	**PROBABILITY AND DISTRIBUTION**	**10 Hours**	3	1,2,3,4
Probability: Definition, sample space, event, probability axioms, conditional probability; Bay's Theorem, Random variable. **Distributions:** Discrete Distribution-Bernoulli theorem, Binomial, and Poisson distribution. Continuous distribution-Normal distribution. Exploring distributions in R. Case study on Stock market analysis.				
UNIT NO.04	**ORDINARY DIFFERENTIAL EQUATIONS**	**10 Hours**	4	1,2,3,4
Definition, Order & Degree of DE., Exact differential equations, Equations reducible to exact equations, Linear differential equations, Reducible to linear differential equations (Bernoulli's equation). **Applications of ODE-** Newton's law of cooling, Law of Decay, Electrical Circuits etc.				
UNIT NO.05	**PARTIAL DERIVATIVES**	**09 Hours**	5	1,2,3,4
Multivariable functions, Introduction to Partial derivatives, Differentiation treating one variable as constant, Composite functions, Total Derivatives, Euler's Theorem on Homogeneous Functions, Applications-Jacobians, functionally dependence/independence.				

Textbooks

1	S. M. Ross, Introduction to probability and statistics for Engineers and Scientists, Elsevier Academic Press, 8th Edition, 2014
2	S. C. Gupta and V. K .Kapoor, Fundamentals of Mathematical Statistics, S. Chand & Sons, 12th edition, 2019.
3	B.V. Ramana, Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.

Reference Books / Journal Articles / Weblink

Reference Books	
1	Ronald E, Walpole, Sharon L. Myers, Keying Ye, Probability and Statistics for Engineers and Scientists, Pearson Prentice Hall, 9th Edition, 2007.
2	Erwin Kreyszig , Advanced Engineering Mathematics, Wiley Publication, 10th Edition, 2017.

List of Tutorials:

Sr. No.	Name of Tutorial
1	Graph It Out: Exploring Data Analysis Through Visual Tools
2	From Data to Decisions: The Role of Mean, mode, median in Analysis of the data
3	The Spread and Shape: Exploring Dispersion and Moments in Statistics using R
4	Analyzing Trends: The Power of Correlation and Regression
5	From Data Points to Smooth Curves: An Introduction to Curve Fitting
6	Analyzing Random Events: A Tutorial on Binomial Distributions
7	Mastering Discrete Distributions :A Tutorial on Poisson Distributions
8	Navigating the Normal Distribution: Properties, Applications, and Insights
9	Probability distributions using R
10	Exact Differential Equations: Signal Dynamics and Control Systems
11	Linear and Bernoulli Equations: Power and Energy Flow Modeling
12	Orthogonal Trajectories and Cooling Laws: Thermal Design in Microchips
13	Multivariable Calculus: Performance Analysis for Multicore Processors
14	Composite and Total Derivatives: Layered Control System Dependencies
15	Homogeneous Functions and Jacobians: Transformation Uses in Robotics

Infosys Springboard Courses

Unit	Topics	Springboard Course	CO
1. Descriptive Statistics	The average of a data set and its properties; the median and mode of a data set and their properties; and the range of a data set and its properties. Then study the inter-quartile range of a data set and its properties; the variance and standard deviation of a data set and their properties	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0135015858483036161485/overview	3
2. Correlation, Regression and curve fitting	Statistics and Probability Distribution	https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138426110813962241879_0_shared/overview	4
3. Probability and Distribution			5
4. Ordinary Differential Equations	Definition, Order & Degree of DE, Exact differential equations, Equations reducible to exact equations (type-1), Equations reducible to exact equations (type-2), Linear differential equations, Reducible to linear differential equations (Bernoulli's equation). Applications of ODE-Orthogonal trajectories, Newton's law of cooling, Law of Decay, Electrical Circuits etc.	Calculus: Getting Started with Derivatives	1
5. Partial Derivatives	Multivariable functions, Introduction to Partial derivatives, Differentiation treating one variable as constant, Composite functions, Chain Rule, Total Derivatives, Euler's Theorem on Homogeneous Functions, Applications, Jacobians, functional dependence/independence	ML Algorithms: Multivariate Calculation & Algorithms	2

BTECES26102: Fundamentals of Electronics

Course Type	ESC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	3Hrs./Week	Lecture	3	CIE Marks	
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	
Practical/Studio	2Hrs./Week	Practical/Studio	1	Total Marks	100
Total	5 Hrs./Week	Total	4		

Course Description

This course will serve as an introduction to develop a foundational understanding of basic electronic components, analogue and linear circuits, instrumentation elements, and communication systems, enabling students to analyse, design, and evaluate simple electronic applications.

Course Outcomes

CO No.	At the end of this course students will be able to …..
1	Identify basic electronic components and devices.
2	Illustrate, explain and demonstrate working of simple electronic circuits based on Diodes, BJTs, FETs
3	Illustrate, explain and demonstrate working of simple Linear Integrated circuits using OP-AMPs.
4	Illustrate, explain and demonstrate working of elements in instrumentation system.
5	Distinguish between different communication systems and evaluate their characteristic requirements.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO3	PSO 4	
CO1	2	1					2					2	1			1	3
CO2	2	1					2					2				1	3
CO3	2	1					2					2				1	3
CO4	1	1					1					2	1			1	3
CO5	1	1					1					2				1	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom’s Taxonomy Level

SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)	
How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?	Integration of AI-Based Tools AI-powered circuit simulation platforms are introduced to allow students to design, test, and analyse electronic circuits virtually. These tools provide real-time feedback, error detection, and performance visualization, helping students understand concepts like rectifiers, amplifiers, and transistor behaviour more effectively. AI-based learning platforms enable students to visualize current flow, voltage variations, and signal behaviour, improving conceptual clarity in topics such as diodes, BJT circuits, and operational amplifiers. Virtual AI labs and simulation environments support hands-on experimentation with physical hardware, enhancing accessibility and reducing dependency on traditional labs.
SECTION B: AI Integration in Teaching & Learning	
How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?	The course incorporates AI-supported simulation and design tools to strengthen the understanding of core electronics concepts through virtual experimentation and real-time analysis. Platforms such as **Tinkercad Circuits** and **Wokwi** are used to design and simulate basic digital and analog circuits in an interactive environment. Advanced circuit analysis is supported using **LTspice** and **Multisim**, enabling students to visualize waveforms, analyze circuit behavior, and validate theoretical concepts. These tools provide a practical approach to learning topics such as rectifiers, transistor circuits, operational amplifiers, and communication systems.
SECTION C: AI Integration in Assessment & Evaluation	
How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?	AI-supported assessment methods are incorporated through the use of virtual lab pre-tests, interactive quiz platforms, and gamified tools such as Kahoot! to evaluate both conceptual understanding and practical application of course content. Virtual lab pre-tests are used for concept clearance before performing experiments, while post-activities assess practical implementation skills. AI-enabled quiz platforms provide instant feedback, performance analytics, and continuous assessment opportunities for students.

Course Content

Unit No.01	**Introduction to Electronics Engineering and Circuits**	**9 Hours**	**CO**	**BTL**
Introduction to different areas of Electronics and their applications in Engineering,, Diode applications- Half-wave rectifier, Full-wave rectifiers (Dual complementary bridge rectifier), Capacitor Filters, Zener diode-Voltage regulator.			1,2	3
Unit No.02	**Bipolar Junction Transistor (BJT) Circuits**	**9 Hours**	1,2	3
BJT types, structure, operation and Characteristics, Transistor in Common-Emitter CE, Transistor as a CE amplifier, Transistor as a switch.				
Unit No.03	**Linear Integrated Circuits**	**9 Hours**	3	3
Introduction to operational amplifiers, Block diagram of OP-AMP, Ideal parameters of OP-AMP, Negative feedback, Inverting & Non inverting Amplifier, Comparators, Summing amplifier, Difference amplifier.				
Unit No.04	**Sensors and Transducers**	**9 Hours**	4	3
Introduction to instrumentation system, Analog and digital data acquisition, Characterstics of Sensor and transducer, Types of sensor and transducer-Active & Passive Sensor, unipolar and bipolar sensor and transducers. RTD , Thermocouple, Strain gauge, LVDT.				
Unit No.05	**Communication system**	**9 Hours**	5	3
Importance of communication system, elements of communication system, bandwidth requirement, IEEE frequency spectrum, Transmission media: Wired (twisted cable, coaxial cable and optical fiber cables) and wireless, Need for modulation, analog modulation schemes: - AM, FM.				

Textbooks

Text Books
1. Floyd, “Electronic Devices and Circuits”, Pearson Education.
2. Robert L. Boylestad, Louis Nashelsky-“Electronic Devices and Circuit Theory” -Prentice Hall
3. R P Jain,” Modern Digital Electronics”, TMH publications
4. H.S. Kalasi “Electronic Instrumentation’, Tata McGraw Hill
5. Frenzel. “Communication Electronics. Principles and Applications.” Third Edition. McGraw Hill Education.

Reference Books / Journal Articles / Weblink

Reference Books
6. Albert Malvino, ”Electronic principles”, 7thEd, TataMc-Graw-Hill.
7. John G. Webster, “The Measurement, Instrumentation & sensors Handbook”, CRC Press.
8. Ramakant Gaikwad, “Linear Integrated Circuits”, Pearson Education.
9. Anand Kumar, ”Fundamental of digital circuits”, prentice Hall of India.

List of Experiments/Studio Assignments, etc

Sr. No	Name of Experiment/ Studio Assignment	CO	BTL
1	Introduction of Electronic Components, Test and measurement instruments.	1	2
2	Observe the given AC voltage and its rectified output , output with capacitor filter of HWR and FWR	1,2	3
3	Zener diode characteristics , Zener as a voltage regulator	1,2	3
4	Single Stage BJT Common Emitter amplifier circuit. For a given BJT CE Amplifier circuit, a. Identify pins of a BJT (such as BC547) and study of its data sheet specifications. b. To measure voltages and observe waveforms at input and output terminals of Single stage BJT Common Emitter amplifier circuit. Calculate voltage gain of the amplifier	2	3
5	BJT as a switch	2	3
6	Op-amp based amplifiers circuits. Identify pins of an Op-amp (such as LM741). Implement Inverting & Non inverting DC Amplifiers.	3	3
7	Op-amp based amplifiers circuits-Summing amplifier and difference amplifier.	3	3
8	Build and Test an Instrumentation system with any one of the following sensors/ Transducers Resistive, Inductive Capacitive	4	3

Infosys Springboard Courses mapped.

Course Name	Contents	Links	CO
Electronics Course	Course Overview: This immersive Electronics course takes participants on a stimulating journey through the essential principles, components, and applications of electronics. Whether you're a novice with a curiosity for circuits or an aspiring engineer seeking to expand your skill set, this course caters to all levels of experience, fostering a strong foundation in electronics.	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01373779399397376019/overview**	**1**
Sensors and Sensor Circuit Design	After taking this course, you will be able to:Understand how to specify the proper thermal, flow, or rotary sensor for taking real-time process data, Implement thermal sensors into an embedded system in both hardware and software,Add the sensor and sensor interface into a microprocessor based development kit,Create hardware and firmware to process sensor signals and feed data to a microprocessor for further evaluation,Study sensor signal noise and apply proper hardware techniques to reduce it to acceptable levels..	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01422273501183180812 81/overview**	**4**
BJT	A bipolar junction transistor is a type of transistor that relies on the contact of two types of semiconductor for its operation. BJTs can be used as amplifiers, switches, or in oscillators. With this course, you will clear all the concepts of bipolar transistors(BJT)	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01384172822159360023 70_shared/overview**	**2**
Semiconductor Electronics:Materials, Devices And Simple Circuits	Semiconductor Electronics:Materials, Devices And Simple Circuits	**https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01396008994203238454/overview**	**1**

BTECES26103: Computer Programming

Course Type	ESC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	3 Hr./Week	Lecture	3	CIE Marks	
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	100
Total	5 Hr./Week	Total	4		

Course Description

It is an introductory programming course for understanding programming concepts and methods using C and C++ programming language.

Course Outcomes

CO No.	Statement
1	Apply programming fundamentals to construct concise solutions.
2	Demonstrate available algorithmic principles to generate efficient solutions.
3	Justify modular programming approach by making use of elementary as well as superior data structures & Implement programming constructs and derived data types to solve real-world problems.
4	Demonstrate the basic object-oriented programming concepts.
5	Apply advanced OOPs concepts in real world applications.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1		3	1										1		2		1,2
CO2	3	2		1	3									1	2		3
CO3	1	3							1					1		1	3,5
CO4	1	2		3									1	3	2	1	3,4
CO5			3	1	2				1							1	3,6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)	
How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?	The course is updated by incorporating the concepts of process oriented and object oriented programming together to make fundamental concepts strong. Students were encouraged to use claude, copilot. AI into C/C++ implementing a **basic machine learning algorithm to make predictions based on similarity from scratch**. This introduces real AI logic of **decision-making based on data** while staying within core programming concepts. This approach used in designing lab assignments to make more "AI-like": ● Basic classification ● Store past student data in arrays ● Compare new input with existing data (basic pattern matching) ● Feature representation ● Distance metrics (Euclidean distance) ● Classification logic ● Training vs testing data ● Improve predictions using simple scoring systems ● Data-driven decisions This method builds a strong bridge between **C/C++ fundamentals and real AI techniques** without relying on external libraries. Such an approach keeps programming within C/C++ fundamentals while introducing AI thinking. It avoids heavy mathematics but builds a strong foundation for future learning in machine learning and data science. Platforms for AY 2026-27: opencv, dlib, ONNX Runtime
SECTION B: AI Integration in Teaching & Learning	
How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?	AI tools such as GitHub Copilot, ChatGPT can enhance the teaching and learning of C and C++ for first-year students by acting as **intelligent learning assistants**. These tools help students understand concepts by giving hints, explaining errors (like pointer mistakes or syntax issues), and showing step-by-step solutions, which improves conceptual clarity. To support **student-centered learning**, these AI tools adapt to individual needs. For example, if a student

	struggles with loops or arrays, the tools can generate simpler practice problems and gradually increase difficulty. They also assist in mini-projects, allowing students to explore coding at their own pace and build confidence. Not as an AI tool but an effective learning platform of hackerrank is suggested for practice. Platforms for AY 2026-27: Replit Ghostwriter, CodeBERT, Tabby, Jupyter Notebook
SECTION C: AI Integration in Assessment & Evaluation	
How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?	To ensure **fairness and transparency**, clear rubrics are shared with students, and AI grading is based on predefined criteria such as output accuracy, code structure, and test case performance. Students can review feedback and request re-evaluation if needed. Inclusivity is maintained by allowing multiple attempts, providing hints, and supporting different learning speeds. For **ethical use**, students are encouraged to use AI only for learning support, not copying solutions. Viva or code explanation sessions help verify understanding. Platforms for AY 2026-27: CodeGrade, Gradescope, and **Tabby (learn), CodeGeex (practice), OpenNN (Implement),** Ok(OKPy), WebCAT, Taskgrader

Course Content

Unit No 1	**Fundamentals of Computers & C Programming**	**Hours 9**	**CO**	**BTL**
Computer Systems & Computing Environments, Computer Languages, Creating & Running Programs, Algorithms & Flowcharts, Introduction to C Language: History, Compiler, Linker, Library & Header Files, Structure of a C Program, Identifiers, Variables, Constants, Data Types, Basic Input/Output, Operators: Arithmetic, Relational, Logical, Bitwise, Expressions, Precedence & Type Conversions, Control Structures: if, switch, goto			1	1,2
Unit No 2	**Loops, Arrays & Strings in C**	**Hours 9**		
Loops: while, for, do-while, break, continue, Arrays: Declaration, Initialization, Memory Layout,			1,2	1,2,3

1D & 2D Arrays, Matrix Operations, Sparse Matrices, Character Arrays (Strings) & String Operations, Menu Driven Programs using Arrays & Strings				
Unit No 3	**Functions, Pointers & Structures in C**	**Hours 9**		
Functions: Anatomy, Local & Global Scope, Nested Calls, Returning Values, Recursive Functions & Types, Pointer Concepts: Definition, Address, Dereferencing, Pointer Arithmetic, Pointers with Functions & Arrays, Structures: Definition, Declaration, Initialization, Array of Structures, Structure within Structure			1,2,3	1,2,4
Unit No 4	**Introduction to C++ & OOP Concepts**	**Hours 9**		
Procedure Oriented vs Object-Oriented Programming, Overview of Identifiers, Variables, Constants, Operators and OOP Principles (Abstraction, Encapsulation, Inheritance, Polymorphism), Introduction to Program Structure, namespace, I/O (cin, cout), Typecasting, type conversion, Control Structures in C++, Classes & Objects: Private, Public, Protected, Members, Encapsulation: Concepts, Benefits, Examples, Static Members and Function, Constructors & Destructors, getter and setter functions and friend function.			1,2,4	1,2,5
Unit No 5	**Advanced OOP Features in C++**	**Hours 9**		
Inheritance: Concept, Modes and Types (Single, Multiple, Multilevel, Hierarchical), Inline Functions vs Macros, Polymorphism: Function Overloading (Compile Time Polymorphism), operator overloading, Virtual Functions (Run-time Polymorphism), Abstraction, Exception Handling (try, catch, throw), Templates: Function & Class Templates			1,4,5	1,2,6

Textbooks

1	"Computer Fundamentals", P. K. Sinha, Sixth Edition, BPB Publication, ISBN 81-7656-752-3.
2	"Let us C", Y. Kanetkar, Second Edition, BPB Publication. ISBN: 8176566217.
3	E. Balagurusamy; "Object oriented programming with C++"; 4thEdition, Tata McGraw-Hill.

Reference Books / Journal Articles / Weblink

1	"Programming language – ANSI C", Brain W Kernighan and Dennis Ritchie, Second edition ISBN 0-13-110370-9
2	"C By Example", Kalicharan Noel, UK, Cambridge University Press.
3	"Programming with C- Schaum's outline Series", B. Gottfried, Second edition, Tata McGraw Hill Publication, ISBN 0-07-463491-7
4	"C and Data Structures", E. Balaguruswamy, TMH Publication, 2003

5	R. Lafore, "The Waite Group's Object oriented Programming in C++"; 3rd Edition, Galgotia Publications.
6	E. Balagurusamy; "Programmmng with Java"; 5thEdition, Tata McGraw-Hill.
7	Bjarne Stroustrup ;" Object-Oriented Programming in C++ " ;4th Edition, Sams Publishing.

List of Experiments/ Studio Assignments, etc.

1	**Number Wizardry — Basic Numeric Operations** Write a C program that accepts a temperature in Celsius and converts it to Fahrenheit and Kelvin. Additionally, find the smallest and largest among five integers, and determine whether a number is positive, negative, or zero.
2	**Matrix Mastery — 2D Array Operations** Write a C program to perform the following on 3x3 matrices: (a) accept and display a matrix, (b) add two matrices, (c) multiply two matrices, and (d) find the transpose of a matrix. Display results in a formatted table.
3	**String Sculptor — User-Defined String Functions** Without using built-in string functions, implement your own: myStrlen, myStrcpy, myStrcat, myStrcmp, and myStrrev. Test each function with sample strings and verify correctness.
4	**Recursion Realm — Factorial & Fibonacci** Write a C program that implements both recursive and non-recursive (iterative) versions of: (a) Factorial of N, and (b) Fibonacci series up to N terms. Demonstrate that both approaches yield identical results.
5	**Pointer Playground — Array of Pointers & Pointer to Array** Write a C program demonstrating: (a) an array of pointers to strings — store 5 student names and sort them alphabetically; (b) a pointer to a 1D integer array — traverse and display elements using pointer arithmetic instead of index notation.
6	**Data Vault — Structures, Unions** Write a C program using a structure Employee (ID, name, salary) to: (a) add records to a binary file, (b) read and display all records, (c) search by employee ID. Also create a union Data to demonstrate memory sharing between int, float, and char members.
7	**Student Register — Class, Objects & Arrays of Objects** Define a class Student with data members: name (string), roll number (int), and grade (char). Create an array of min. 3 Student objects. Write member functions to read data from the user and display all student details in a formatted table.
8	**Inheritance Tower — Single & Multilevel Inheritance** Write a C++ program to illustrate: (a) single-level inheritance — base class Vehicle with derived class Car; (b) multilevel inheritance — Animal → Mammal → Dog chain. Each class adds new attributes and methods. Demonstrate access to inherited members.
9	**Safety Net — Exception Handling with Multiple Catch** Write a C++ program that performs integer division. Use a try block to detect: (a) division by zero (throw int), (b) negative operand (throw string), and (c) a generic error (catch-all). Each catch block displays a descriptive error message. Show execution continues after handling.
10	**Generic Engine — Template Class** Implement a template class Stack<T> that supports push(), pop(), peek(), and isEmpty() operations. Instantiate the stack for (a) integers, (b) floats, and (c) strings, demonstrating that the same class definition works across all data types.
11	Evaluation / Hackathon on Assignment 1 & 2
12	Evaluation / Hackathon on Assignment 3 & 4
13	Evaluation / Hackathon on Assignment 5 & 6
14	Evaluation / Hackathon on Assignment 7 & 8
15	Evaluation / Hackathon on Assignment 9 & 10

Springboard courses:

Unit	Topics	Springboard Course	CO
1. Fundamentals of Computers & C Programming	Computer Systems & Computing Environments, Computer Languages, Creating & Running Programs, Algorithms & Flowcharts, Introduction to C Language: History, Compiler, Linker, Library & Header Files, Structure of a C Program, Identifiers, Variables, Constants, Data Types, Basic Input/Output, Operators: Arithmetic, Relational, Logical, Bitwise, Expressions, Precedence & Type Conversions, Control Structures: if, switch, goto	**C Programming 101** **C Programming Course** **Programming in C** **Fundamentals of the C Language: Variables & Datatypes**	**1,2**
2. Loops, Arrays & Strings in C	Loops: while, for, do-while, break, continue, Arrays: Declaration, Initialization, Memory Layout, 1D & 2D Arrays, Matrix Operations, Sparse Matrices, Character Arrays (Strings) & String Operations, Menu Driven Programs using Arrays & Strings	**C Programming 101** **C Programming Course** **Handling Arrays in C Programming** **Two-Dimensional Arrays and Pointers in C Programming in C**	**1,2,3**
3. Functions, Pointers & Structures in C	Functions: Anatomy, Local & Global Scope, Nested Calls, Returning Values, Recursive Functions & Types, Pointer Concepts: Definition, Address, Dereferencing, Pointer Arithmetic, Pointers with Functions & Arrays, Structures: Definition, Declaration, Initialization, Array of Structures, Structure within Structure	**C Programming Course** **Working with Functions in C Program** **Programming in C** **Pointers In C Programming** **Two-Dimensional Arrays and Pointers in C**	**2,3**

4. Introduction to C++ & OOP Concepts	Procedure Oriented vs Object-Oriented Programming, Overview of Identifiers, Variables, Constants, Operators and OOP Principles (Abstraction, Encapsulation, Inheritance, Polymorphism), Introduction to Program Structure, namespace, I/O (cin, cout), Typecasting, type conversion, Control Structures in C++, Classes & Objects: Private, Public, Protected, Members, Encapsulation: Concepts, Benefits, Examples, Static Members and Function, Constructors & Destructors, getter and setter functions and friend function.	**C++ Fundamentals**	**4, 5**
5. Advanced OOP Features in C++	Inheritance: Concept ,Modes and Types (Single, Multiple, Multilevel, Hierarchical), Inline Functions vs Macros, Polymorphism: Function Overloading (Compile Time Polymorphism), operator overloading, Virtual Functions (Run-time Polymorphism), Abstraction, Exception Handling (try, catch, throw), Templates: Function & Class Templates	**Programming using C++**	**4, 5**

Projects on Infosys Springboard:

1. Project in C Programming - Hangman Game
2. Project on C Programming - Payroll Processing
3. Project in C Programming - Employee Management System
4. Project in C Programming - 2 Players Tic-Tac-Toe Game

BTECES26104: Web Application Development

Course Type	VSEC	Semester	1

Teaching Scheme		Credits		Examination Scheme	
Lecture	1 Hr./Week	Lecture	1	CIE Marks	
Tutorial	-- Hr./Week	Tutorial	-	ESE Marks	
Practical/Studio	2 Hr./Week	Practical/Studio	1	Total Marks	100
Total	3 Hr./Week	Total	2		

Course Description

This course helps the student to understand the principles and methodologies of web-based application development process and current client-side scripting technologies

Course Outcomes

CO No.	At the end of this course students will be able to …..
1	Analyze given assignment to select sustainable web development, design methodology
2	Develop web-based application using suitable client-side web technologies
3	Enable to design and develop solutions to problems using technologies studied.
4	Design and develop solution to handle information which is location independent.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PS O3	PSO 4	
CO1	3																1,2
CO2				3													1,2,3
CO3							3										1,2,3, 6
CO4													3				1,2,3, 6

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)	
How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?	**Unit 1** students are introduced to the role of AI in the evolution of the web, including its application in search engines, browsers, and intelligent web systems, along with discussions on ethical considerations such as bias, privacy, and responsible usage. **Unit 2** incorporates AI-driven design concepts, where students explore how AI tools assist in generating layouts, color schemes, and improving accessibility in web interfaces. **Unit 3** focus shifts to practical implementation, where students learn to integrate AI functionalities using JavaScript and REST APIs, including the development of simple chatbot interfaces and AI-assisted form validation. **Unit 4** extends these concepts to real-world applications by introducing AI in e-commerce systems, such as recommendation engines, personalized user experiences, and automated customer support. The course has been further strengthened by incorporating selected modules from Infosys Springboard as supplementary learning resources. These courses provide structured exposure to foundational concepts in Artificial Intelligence, web technologies, and digital skills, which complement the core syllabus. ◇ **GitHub Copilot** ● Suggests HTML, CSS, JavaScript code automatically ● Coding Tutor ◇ **Replit Ghostwriter** ● Runs in browser (no installation needed) ● AI helps write, debug, and explain code ● Best for labs and beginner ◇ **Codeium**

	● Freely available ● Includes chat + code explanation
SECTION B: AI Integration in Teaching & Learning	
How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?	Beginner-friendly tools such as ChatGPT, Replit, and Figma are incorporated to facilitate interactive and hands-on learning experiences. These tools assist students in generating code, visualizing designs, and understanding complex concepts through guided exploration. Coding Copilot, Codeium, Replit Design Figma, Wizard Website Hostinger AI Builder AI Interaction ChatGPT, Gemini The course adopts a project-based and activity-oriented approach, encouraging students to build simple web applications integrated with AI features such as chat interfaces or recommendation sections. Flipped classroom techniques are also employed, where AI-generated summaries and explanations are used as pre-learning material, allowing classroom time to focus on discussion, problem-solving, and application.
SECTION C: AI Integration in Assessment & Evaluation	
How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?	Infosys Springboard is also utilized as a supportive tool in assessment and evaluation. Completion of relevant modules, quizzes, and certifications from Infosys Springboard are considered as part of continuous assessment. Mini-projects are introduced where students develop simple web applications incorporating AI features such as chatbots or basic recommendation systems, thereby demonstrating practical integration skills. Assessment Tools: Mentimeter, Kahoot, Quiz.com Mini Projects: Cloning of Website, Poster Presentations

Course Content

Unit No 1	**Introduction & HTML**	**Hours 4**
Evolution of Web, Web Browser, Web Server, Application server, Client Server Architecture and Communication, Client Server communication techniques, URL, URI, web design concepts and issues, HTTP request response structure of html document, elements: headings, paragraphs, line break, colors& fonts, links, frames, lists, tables, images and forms, page division using div flex		
Unit No 2	**Cascading Style Sheets**	**Hours 3**
DTD: Schema, elements, attributes, Benefits, syntax, inline, internal, external style sheets, CSS selectors (id, class, grouping, child), display behavior and positioning (static, fixed, relative, absolute, floats),visibility of element attribute, adding scrollbars, Modern CSS (Flexbox, Grid)		
Unit No 3	**JavaScript & Server-Side Scripting**	**Hours 5**
HTML (Embedded, External), variable declarations, operators, Control statements, Arrays, Functions and Scopes, Objects in Java Scripts, regular expressions for input validations, error handling DOM: DOM levels, DOM Objects and their properties and methods, Manipulating DOM Introduction to Server-Side Programming, Core Concepts: HTTP, Request/Response, RESTful APIs,Server		
Unit No 4	**E-commerce Web Application Development**	**Hours 3**
Introduction to E-commerce (Types, Models: B2C, B2B, C2C),Essential E-commerce Features (Catalog, Cart, Checkout, Payment Integration), UI/UX Principles for Web Applications		

Textbook

1	*Achyut Godbole & Atul Karate, Web Technologies: TCP/IP to Internet Application Architectures‖, McGraw Hill Education publications, ISBN, 007047298X, 9780070472983*
2	*Ralph Moseley & M. T. Sav Aliya, ―Developing Web Applications, Wiley publications, ISBN 13: 9788126538676*

Mini Project

1. Students have to implement mini project which should include following concepts: HTML, CSS, JavaScript

List of Practice Assignments:

2. Design web pages which will include concepts of HTML
3. Apply CSS to the created web pages.
4. Write java script where ever required for web pages created or for input validation

Suggestions:

1. Read and Display syllabus structure, add filter option
2. Design page to get student details information with validation
3. Design pages to display food information with images

Unit	Topics	Springboard Course	CO
Introduction & HTML	Evolution of Web, Web Browser, Web Server, Application server, Client Server Architecture and Communication, Client Server communication techniques, URL, URI, web design concepts and issues, HTTP request response structure of html document, elements: headings, paragraphs, line break, colors& fonts, links, frames, lists, tables, images and forms, page division using div flex	**What's the World Wide Web (WWW) And How it Works** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013301321289031680791 1_shared/overview **Introduction to HTML** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013841749346058240288 1_shared/overview **HTML Begineers** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013843238343647232339 59_shared/overview **HTML Tutorial** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_014235406833754112316 6/overview	**1**
Cascading Style Sheets	DTD: Schema, elements, attributes, Benefits, syntax, inline, internal, external style sheets, CSS selectors (id, class, grouping, child), display behavior and positioning (static, fixed, relative, absolute, floats),visibility of element attribute, adding scrollbars, Modern CSS (Flexbox, Grid)	**Learning CSS** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013094432869507072228 6_shared/overview **Practical HTML and CSS** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013303869999046656285 99_shared/overview	**2**
JavaScript & Server-Side Scripting	HTML (Embedded, External), variable declarations, operators, Control statements, Arrays, Functions and Scopes, Objects in Java Scripts, regular expressions for	**Javascript** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_18109698366332810000_shared/overview	**3**

	input validations, error handling DOM: DOM levels, DOM Objects and their properties and methods, Manipulating DOM Introduction to Server-Side Programming, Core Concepts: HTTP, Request/Response, RESTful APIs,Server	**Javascript Tutorial** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_014222738523209728949/overview	
E-commerce Web Application Development	Introduction to E-commerce (Types, Models: B2C, B2B, C2C),Essential E-commerce Features (Catalog, Cart, Checkout, Payment Integration), UI/UX Principles for Web Applications	**Introduction to Web Application Development** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013842618018390016188_90shared/overview **Introduction To UI/UX** https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013501558940262400877 3/overview	**4**

BTECES26105: Applied Science and Engineering Project I

Course Type	BSC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	04		

Course Description

This course is focusing on understanding the core principles and mindset that drive design thinking. This course will be used to complete the design thinking process to tackle real-world challenges and discover its business applications by creative and critical thinking.

Course Outcomes

CO No.	Statement
1	**CO1:** Identify and define real-world problems through stakeholder analysis, literature review, and benchmarking across societal and scientific domains.
2	**CO2:** Develop and implement solution strategies—products, systems, or experiments—using appropriate tools, resources, and interdisciplinary knowledge.
3	**CO3:** Communicate and refine project outcomes through iterative prototyping, documentation, peer feedback, and structured project planning (e.g., Gantt charts, risk analysis).

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	1	1	3			1	1	1	2	1	1	1					1,2,3
CO2						2	2	1	1	2	1	1					2,3
CO3							3			1		2					1,2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

<table>
<tr><td colspan="2">SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)</td></tr>
<tr><td>How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?</td><td>1. Literature Review (Understanding, not overload)

● SciSpace
○ Paste a PDF → get simple explanations
○ Good for beginners who find papers difficult
● Elicit
○ Ask questions like “What is machine learning?”
○ Gives summarized research insights</td></tr>
<tr><td colspan="2">SECTION B: AI Integration in Teaching & Learning</td></tr>
<tr><td rowspan="2">How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?</td><td>2. Writing (Reports, Assignments, Mini Projects)

● ChatGPT
○ Generate structure (Introduction, Method, Conclusion)
○ Rewrite in simple academic language
● QuillBot
○ Improve sentences and avoid copying</td></tr>
<tr><td>3. Grammar & Clarity (Very important for beginners)

● Grammarly
○ Fix grammar mistakes
○ Improve sentence clarity</td></tr>
<tr><td colspan="2">SECTION C: AI Integration in Assessment & Evaluation</td></tr>
<tr><td>How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?</td><td>4. Documentation / Project Reports

● Notion AI
○ Organize notes, reports, and project ideas
○ Easy to maintain structured documentation

Use this 4-step process:
Step 1: Understand topic
● Use SciSpace or Elicit</td></tr>
</table>

	● Goal: "What is this topic?" **Step 2: Make rough notes** ● Write key points in your own words **Step 3: Draft report** ● Use ChatGPT to: ○ Create structure ○ Expand points into paragraphs **Step 4: Improve writing** ● Use QuillBot + Grammarly ● Finalize clean report
	Example (Mini Project Flow) Topic: *Calculator using Python* 1. Understand → "What is a calculator program?" 2. Notes → Input, operations, output 3. Draft → Introduction + Code explanation 4. Improve → Grammar + formatting
	Important Guidance to be given ● Do **not copy directly from AI** ● Always **rewrite in your own words** ● Use AI as a **helper, not a replacement** ● Focus on **understanding + basic writing skills** **Final Recommendation (Keep it minimal)** If students use just these 3, it is enough: ● **SciSpace** → understanding ● **ChatGPT** → writing ● **Grammarly** → correction

COURSE CONTENT

UNIT NO. 01	Project Execution Guidelines	30 Hours	CO	BTL
1.	Introduction to design and scientific problem-solving		CO1 - 1 to 7	1,2,3
2.	Brainstorming issues across domains: environment, health, energy, mobility, etc.			
3.	Team/individual project selection and Stakeholder identification (e.g., users, community, clients)		CO2 – 8 to 11	
4.	Literature review and benchmarking			
5.	Literature review and benchmarking - continue			
6.	Literature review and benchmarking			
7.	Creative and collaborative problem solving – open ended discussion		CO3- 9 to	

8. Finalizing the problem scope with constraints and impact and translate problem into clear objectives or testable hypotheses 9. Choose solution format: product, process, system, app, experiment and identify needed knowledge/skills (coding, circuits, design, math, etc.) 10. Learn selected tools (e.g., Python, Arduino, CAD, Excel, Power BI, ML tools) 11. Resource planning: materials, software, tools 12. Prepare timelines using Gantt charts and Risk analysis and contingency planning 13. Initial prototype building (physical/software) or setting up experiments 14. Iterative refinement: making small changes based on observations 15. Documentation of technical process and Peer feedback on prototype/experiment setup	15	

Textbooks

1	

Reference Books / Journal Articles / Weblink

Reference Books	
1	https://www.linkedin.com/posts/kamal-shahid-0126_read-10-papers-in-1-hour-activity-7346830717152022528-LWeQ?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA https://www.linkedin.com/posts/ali-mk-hindi-%F0%9F%87%AE%F0%9F%87%B6%F0%9F%87%B5%F0%9F%87%B8-4b1b80115_all-the-research-tools-you-need-in-2025-activity-7341087359205007362-8kdX?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA https://www.linkedin.com/posts/drasadnaveed_need-help-choosing-a-new-research-topic-activity-7347110335511674880-MU8u?utm_source=share&utm_medium=member_android&rcm=ACoAABnxYZ8BfhWRcVTlUyaeChOpLEy5VudBSDA

BTECES26106: Value Added Career Orientation I

Course Type	CC	Semester	I

Teaching Scheme		Credits		Examination Scheme	
Lecture	02Hr./Week	Lecture	02	CIE Marks	
Tutorial	00Hr./Week	Tutorial	00	ESE Marks	
Practical/Studio	00Hr./Week	Practical/Studio	00	Total Marks	100
Total	02Hr./Week	Total	02		

Course Description

To empower F.Y.B.Tech students with career readiness, soft skills, and holistic personal development through collaborative exploration of resources, group projects, and team-based creative assessments.

Course Outcomes:

Upon completion of this course, the student will be able to:

CO No.	Statement
1	Reflect on your strengths, interests, and learning style, and begin to plan your academic and career journey
2	Identify different types of careers, work models, and new trends like remote work, gig economy, and use of technology in jobs.
3	Understand how to work with people from different cultures and industries and learn basic teamwork and workplace skills.
4	Explore how new ideas are developed through research and innovation and understand the basics of entrepreneurship.
5	Recognize the importance of giving back to society, volunteering, and learning about careers in government and social sectors.

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4	BTL
CO1								1	1			3		1		1	4
CO2					1	1					2				1	1	1
CO3							1	1	1		1			1			2
CO4						3	2			2			1		1		3
CO5							1	1		3	2				1	1	2, 4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)	
How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?	● Introduction to AI awareness and digital readiness for career development. ● Use of AI tools for career exploration, resume guidance, and skill mapping. ● Exposure to AI's role in future careers, entrepreneurship, and workplace transformation. ● Simple case studies on startups, emerging careers, and digital work environments. ● Focus on balancing technology use with human values and interpersonal skills.
SECTION B: AI Integration in Teaching & Learning	
How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?	● AI used as a supplementary aid for career planning and skill development. ● Faculty-led discussions, mentoring, teamwork, and reflective activities remain central. ● Moderate use of AI tools for resume review, interview practice, and communication support. ● Students encouraged to critically evaluate AI suggestions. ● Emphasis on creativity, ethics, leadership, empathy, and collaboration.
SECTION C: AI Integration in Assessment & Evaluation	
How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?	● Assessments include reflections, presentations, group activities, and case studies. ● AI tools may be used selectively for research and idea generation. ● Greater emphasis on originality, communication, teamwork, and personal growth. ● Faculty evaluation through rubrics and observation ensures fairness and authenticity. ● Course outcome attainment measured through participation and skill development.

COURSE CONTENT

UNIT NO. 01	LEARNING STRATEGIES & ACADEMIC PROGRESSION	06 Hours	CO	BTL
Self-awareness- values, interests, personality, and purpose; Career-aligned academic planning (UG+PG, India & abroad); Learning how to learn- Growth mindset, metacognition, self-regulation; Building transferable skills- Communication, critical thinking, collaboration, digital fluency; Navigating interdisciplinary learning and lifelong learning paths			1	1, 2
UNIT NO.02	**CAREER ORIENTATION & FUTURE OF WORK**	**06 Hours**		
Defining a career; Career vs. job vs. livelihood; Evolving work models- Portfolio careers, gig economy, hybrid & remote work; Career success in the 21st century- Purpose, flexibility, and impact; Leveraging technology- AI, automation, and digital transformation; Interdisciplinary thinking and ESG awareness in shaping careers			2	2, 3
UNIT NO.03	**GLOBAL CITIZENSHIP & INDUSTRY ENGAGEMENT**	**06 Hours**		
Global citizenship and intercultural fluency; Working in global teams and across cultures; Teamwork, collaboration, conflict resolution, and inclusion; Introduction to business sectors and industries; Understanding corporate etiquette and workplace dynamics			3	3, 4
UNIT NO.04	**INNOVATION & ENTREPRENEURSHIP**	**06 Hours**		
Curiosity and problem-solving through research; Innovation ecosystems- Labs, incubators, hackathons; Entrepreneurial mindset and social enterprise; From ideas to action: Generating, refining, pitching; Learning from real-world examples- Startup or innovation journeys			4	2, 3
UNIT NO.05	**COMMUNITY ENGAGEMENT & GOVERNMENT/PUBLIC SECTOR**	**06 Hours**		
Civic responsibility and active citizenship; Volunteerism and service-learning; Understanding social innovation and grassroots impact; Engaging with local communities and marginalized voices; Public policy awareness and sustainable development goals (SDGs)			5	4, 5

Reference Books:

1.	“What Color Is Your Parachute? for College: Pave Your Path from Major to Meaningful Work” by Katherine Brooks, Ten Speed Press (2021).
2.	"The Start-up of You: Adapt, Take Risks, Grow Your Network" by Reid Hoffman and Ben Casnocha, Random House Business (2013).
3.	“Creative Confidence: Unleashing the Creative Potential Within Us All” by Tom Kelley and David Kelley (2013).

Infosys Springboard Courses:

Unit	Topics	Springboard Course	CO
Learning Strategies & Academic Progression	Metacognition, self-regulation, study habits	Learning How to Learn https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01350156510110515 29522/overview	1
Career Orientation & Future of Work	Industry trends, automation, tech impact	Future of Work https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01435729566264524 837/overview	2
Global Citizenship & Industry Engagement	Collaboration, cross-cultural communication, inclusion	Interpersonal Skills https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01389695652828774 4589_shared/overview	3
Innovation & Entrepreneurship	Small business planning and strategy	Entrepreneurship-Introduction to Business Model Canvas https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01384347494667878 439748_shared/overview	4
Community Engagement & Government/Public Sector	Innovation	Innovation https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01378218558860492 8167/overview	5

IKS25AI01A: Indian Knowledge System (IKS)

Course Type	IKS	Semester	1

Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hrs./Week	Lecture	2	CIE Marks	
Tutorial	0 Hrs./Week	Tutorial	0	ESE Marks	
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	100
Total	2 Hrs./Week	Total	2		

Course Discription

The purpose of the course is to provide a general introduction to Indian Knowledge System (IKS) and sensitize the students to the contributions made by ancient Indians in the field of Science, Philosophy and related applications and concepts.

Course Outcomes

CO No.	At theend of this course students will be able to …..
1	Understand the basic of IKS and Vedic corpus.
2	Understand Mathematical Sciences without knowing sciences whereas, engineering.
3	Understand Astronomy and Engineering technology in metals and metal working
4	Understand the need for an interface. Compare and contrast memory mapping and IO mapping techniques. Understand the Applications in Engineering and Technology
5	Understand Language , Knowledge Framework and linguistic

Mapping of COs to POs and PSOs

CO Number	POs												PSOs				BTL
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4	
CO1	1					2						3	1				,3
CO2		2				1						3	1				2,3
CO3			1			2						3	1				2
CO4			1			2						3	1				2
CO5			1			2						3	1				1,2

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

SECTION A: AI Integration in Course Syllabus (AI-Based Content & Case Studies)	
How has the course been updated to incorporate AI, including the integration of AI-based concepts or tools, and the introduction of relevant AI case studies?	**Concept:** Introduction to **Knowledge Representation**. Mapping the hierarchical organization of the Vedic corpus to modern **Knowledge Graphs** and Ontology engineering. **Case Study:** *The Digital Preservation of Manuscripts* – Using **Computer Vision** and **Generative AI** to restore and digitize damaged palm-leaf manuscripts. **Concept:** Piṅgala's binary system as a precursor to **Boolean Algebra** and modern computer architecture. Introduction to **Hashing Algorithms** using the Kaṭapayādi system. **Case Study:** *Binary Combinatorics in AI* – How ancient Indian prosody (Chandas) informs modern error correcting codes. **Concept:** Using **Machine Learning (Materials Informatics)** to analyze the chemical properties of Wootz steel and the corrosion resistance of the Delhi Iron Pillar. **Case Study:** *Virtual Jantar Mantar* – Using **Augmented Reality (AR)** to simulate astronomical observations. **Concept:** Applying **Generative Adversarial Networks (GANs)** to create floor plans based on Vāstu-śāstra principles and optimized town planning. **Case Study:** *Digital Twin of Hampi* – Using **LiDAR and AI-driven 3D reconstruction** to preserve stone architecture. **Concept:** Aṣṭādhyāyī as the blueprint for **Context-Free Grammars (CFG)**. Exploring the "Rule-Based" vs. "Statistical" approaches in **Natural Language Processing (NLP)**. **Case Study:** *Project Bhashini* – The use of Sanskrit-inspired logic in modern Indian language translation models.
SECTION B: AI Integration in Teaching & Learning	
How are AI tools and strategies used to enhance student understanding, support student-centered learning practices, and ensure ethical and inclusive use of AI in the teaching-learning process?	● **Multilingual Access (Inclusivity):** Use of AI tools like **Anuvadini** and **Bhashini** to provide real-time translation of technical IKS concepts into regional languages, ensuring no student is left behind due to language barriers. ● **AI-Powered Learning Assistants:** Integration of custom-trained **LLM chatbots** (e.g., a "Panini-

	GPT") that allow students to interact with Pāṇinian grammar rules or Nyāya logic through conversational queries. ● **Virtual Laboratories:** Use of AI-simulated environments where students can "construct" Vedic altars (Śulba-sūtras) or test the fluid dynamics of ancient Indian shipbuilding using **Digital Twins**. ● **Ethical Frameworks:** Dedicated sessions on **AI Ethics** through the lens of *Dharma* and *Nyāya* logic, teaching students how to identify and mitigate **Algorithmic Bias** by applying Indian epistemological frameworks.
SECTION C: AI Integration in Assessment & Evaluation	
How are assessments redesigned to incorporate AI for evaluating outcomes, which AI tools are used for assessment and grading, how fairness, transparency, and inclusivity are ensured in AI- supported assessments, and how the impact of these AI-enabled assessments on student performance and course outcome attainment is measured?	**Redesigned Assessments:** Transition from rote-memory exams to **AI-Collaborative Projects**. For example, students are tasked with writing a Python script to automate a specific Sanskrit grammar rule (Pāṇini's rules) and then evaluated on their logic. **Tools for Grading:** ● **AI-Assisted Rubrics:** Using AI to provide preliminary feedback on student essays regarding IKS philosophy, which is then verified by the faculty to ensure transparency. ● **Automated Logic Validation:** Using symbolic AI tools to grade students' mastery of Vedic mathematics and logical syllogisms. **Ensuring Fairness & Integrity:** ● **Proctoring & Watchdogs:** Use of **AI-driven "Watchdog" systems** that flag academic misconduct while maintaining student privacy. ● **Transparency:** All AI-graded components are accompanied by an "Explainable AI" (XAI) report, showing the student exactly how their score was calculated. **Measuring Impact:** Continuous monitoring of **Course Outcome (CO) attainment** through AI analytics, identifying learning gaps in real-time and allowing for adaptive instructional changes mid-semester.

Course Content

Unit No.01	Indian Knowledge System – An Introduction and Vedic Corpus	6 Hours	CO	BTL
IKS Introduction,Organization of IKS,Historicity of IKS,Some salient aspects of IKS, Introduction to Vedas,A synopsis of the four Vedas,Sub-classification of Vedas,Messages in Vedas,Introduction to Vedāṅgas Prologue on Śikṣā and Vyākaraṇa,Basics of Nirukta and Chandas,Introduction to Kalpa and Jyotiṣa,Vedic Life: A Distinctive Features			1	2
Unit No.02	**Number Systems and Units of Measurement and Mathematics**	**6 Hours**	**2**	**2.3**
Number Systems and Units of Measurement: Number systems in India - Historical evidence,Salient aspects of Indian Mathematics,Bhūta-Saṃkhyā system,Kaṭapayādi system,Measurements for time, distance, and weight Piṅgala and the Binary system **Mathematics:** Introduction to Indian Mathematics,Unique aspects of ndian,Mathematics Indian Mathematicians and their Contributions Algebra Geometry Trigonometry Binary mathematics and combinatorial problems in Chandaḥ Śāstra Magic squares in India				
Unit No.03	**Astronomy and Engineering and Technology: Metals and Metalworking:**	**6 Hours**	**3**	**2**
Astronomy: Introduction to Indian astronomy,Indian contributions in astronomy,The celestial coordinate system.,Elements of the Indian calendar Notion of years and months,Pañcāṅga – The Indian calendar system, Astronomical Instruments (Yantras),Jantar Mantar of Rājā Jai Singh Sawai **Engineering and Technology: Metals and Metalworking:** Wootz Steel: The rise and fall of a great Indian technology,The Indian S & T heritage ,Mining and ore extraction,Metals and metalworking technology , Iron and steel in India ,Lost wax casting of idols and artefacts,Aparatuses used for extraction of metallic components				
Unit No.04	**Engineering and Technology: Other applications and Town Planning and Architecture:**	**6 Hours**	**4**	**2**
Engineering and Technology: Other applications: Irrigation systems and practices in South India , Literary sources for science and technology ,Physical structures in India ,Irrigation and water management, Dyes and painting technology, The art of making perfumes Surgical techniques , Shipbuilding , Sixty-four art forms (64 Kalās) , Status of Indigenous S & T. **Town Planning and Architecture:** Perspective of Arthaśāstra on town planning ,Vāstu-śāstra – The science of architecture ,				

Eight limbs of Vāstu ,Town planning ,Temples in India: marvelous stone architecture for eternity , Temple architecture in India Iconography				
Unit No.05	**Knowledge Framework and classifications and Linguistics**	**6 Hours**	**5**	**1,2**
Knowledge Framework and classifications: Indian scheme of knowledge , The knowledge triangle , Prameya – A vaiśeṣikan approach to physical reality , Dravyas – the constituents of the physical reality , Attributes – the properties of substances and Action – the driver of conjunction and disjunction , Sāmānya, viśēṣa, samavāya Pramāṇa – the means of valid knowledge ,Saṃśaya – ambiguities in existing knowledge , Framework for establishing valid knowledge Deductive or inductive logic framework, Potential fallacies in the reasoning process ,Siddhānta: established tenets in a field of study **Linguistics** Introduction to Linguistics , Aṣṭādhyāyī ,Phonetics , Word generation Computational aspects , Mnemonics , Recursive operations , Rule based operations , Sentence formation , Verbs and prefixes , Role of Sanskrit in natural language processing				

Textbooks:

1	**Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavana R.N. (2022),** "Introduction to Indian Knowledge System: Concepts and Applications", PHI Learning Private Ltd. Delhi.

Reference Books / Journal Articles / Weblink

Reference Books	
1	Pride of India: A Glimpse into India's Scientific Heritage, Samskrita Bharati, New Delhi.
2	Sampad and Vijay (2011). "The Wonder that is Sanskrit", Sri Aurobindo Society, Puducherry.
3	Bag, A.K. (1979). Mathematics in Ancient and Medieval India, Chaukhamba Orientalia, New Delhi.
4	Datta, B. and Singh, A.N. (1962). History of Hindu Mathematics: Parts I and II, Asia Publishing House, Mumbai.
5	Kapoor Kapil, Singh Avadhesh (2021). "Indian Knowledge Systems Vol – I & II", Indian Institute of Advanced Study, Shimla, H.P.
6	Subbarayappa, B.V. and Sarma, K.V. (1985). Indian Astronomy: A Source Book, Nehru Centre, Mumbai